MUHAMMAD FAAIZ ALAM

Software Engineering Student | Backend Developer

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PROFESSIONAL SUMMARY

Software Engineering undergraduate (CGPA 3.71) building backend systems with FastAPI and PostgreSQL. Experience in RBAC, blockchain-based credential verification, and AI/RAG integration through an internship, an industry project, and personal projects. Looking to grow toward software architecture by building scalable, well-designed systems.

EDUCATION

NED University of Engineering and Technology Karachi, Pakistan
2024 – 2028
Bachelor of Engineering in Software Engineering
CGPA: 3.709/4.00
Relevant Coursework: Software Design and Architecture, Software Quality Engineering, Database Management Systems, Object-Oriented Programming, Computer Networks, Operating Systems

PROFESSIONAL EXPERIENCE

Software Engineering Intern – IMD, NED University *Internship*
Project: PC Preventive Maintenance Dashboard

- Developed backend services using FastAPI and Windows monitoring agents to collect CPU, RAM, disk, and system health metrics.
- Implemented HTTP-based agent-server communication and automatic agent service management through a persistent launcher.
- Implemented Role-Based Access Control (RBAC) with 3 user roles (IT Manager, IT Support, IT Staff).
- Automated Excel-based maintenance reporting and integrated software inventory and license expiry tracking.
- Optimized agent reporting frequency and introduced application-level caching to reduce unnecessary network traffic and improve dashboard responsiveness across approximately 60 PCs.
- Configured the system on a local laboratory network and tested it across approximately 60 PCs, including Ubuntu environments.

INDUSTRY PROJECT

MIYAAR – Blockchain-Based National Skills Passport *Backend Developer*
Technology: FastAPI, PostgreSQL, Polygon, AWS KMS

- Developed FastAPI backend services for credential issuance, approval, and verification workflows.
- Integrated Polygon blockchain to anchor SHA-256 credential hashes, keeping personal data off-chain in encrypted PostgreSQL.
- Implemented role-based access control, MFA, and AWS KMS signing for secure credential approval.
- Supported DApp-based verification so employers worldwide can instantly confirm credential validity.

SELECTED PROJECTS

StackPilot – AI-Powered Project Management & SDLC Automation Platform
FastAPI, React, PostgreSQL, Gemini API, RAG, JWT, Vercel, Railway

- Developing an AI-powered platform to automate key SDLC activities, including project planning, documentation, and workflow management.
- Integrated Gemini API and a RAG pipeline to support automated generation of Software Requirements Specification (SRS) documents and UML diagrams.
- Implemented Role-Based Access Control (RBAC) for Project Owners, Project Managers, and Editors.
- Developed backend APIs, JWT authentication, and OTP-based email verification for secure user access and project management.
- Deployed on Vercel and Railway with automatic CI/CD via GitHub on every push.

RedBone – Restaurant Ordering Application
React, Node.js, Express.js, MongoDB, Nodemailer

- Developed a food ordering application featuring food cards and basic ordering functionality.
- Integrated Nodemailer to send email notifications when customers place orders.

EcoGuard – Forest Ranger Management System
MERN Stack, Node.js, WebSockets

- Developed a forest ranger management system to support ranger operations and administrative activities.
- Implemented real-time chat functionality using WebSockets and Role-Based Access Control (RBAC).

TECHNICAL SKILLS

Languages: Python, JavaScript, C++
Backend: FastAPI, Node.js, Express.js, REST APIs, MVC Architecture
Frontend: React.js, HTML, CSS, Tailwind CSS
Databases: PostgreSQL, MongoDB, SQL
AI Integration: Gemini API, RAG, Embeddings
Cloud & Deployment: AWS (KMS), Vercel, Railway, CI/CD (GitHub auto-deploy)
Blockchain: Polygon, DApp Integration
Tools: Git, GitHub, Postman, UiPath

CERTIFICATIONS & ACHIEVEMENTS

Google AI Professional Certificate *Google – Coursera*
May 2026 | 7 courses completed

Introduction to FastAPI and Backend Development Fundamentals *Coursera – Packt*
August 2026 | Grade: 90.80%

Legacy Responsive Web Design *freeCodeCamp*
MONTH YEAR

Teaching Assistant – Problem Solving Bootcamp *September – October 2025*
Guided students in programming fundamentals and problem-solving.